

ALTERNATING CURRENTS

An understanding of the practical and economic advantages of transmission of power by electricity from Cambridge IGCSE / O Level Physics or equivalent is assumed.

21.1 Characteristics of alternating currents

Candidates should be able to:

- 1 understand and use the terms period, frequency and peak value as applied to an alternating current or voltage
- 2 use equations of the form $x = x_0 \sin \omega t$ representing a sinusoidally alternating current or voltage
- 3 recall and use the fact that the mean power in a resistive load is half the maximum power for a sinusoidal alternating current
- 4 distinguish between root-mean-square (r.m.s.) and peak values and recall and use $I_{\text{r.m.s.}} = I_0 / \sqrt{2}$ and $V_{\text{r.m.s.}} = V_0 / \sqrt{2}$ for a sinusoidal alternating current

1. O/N/09/41/Q7

A sinusoidal alternating voltage is to be rectified.

- (a) Suggest one advantage of full-wave rectification as compared with half-wave rectification.

.....
..... [1]

- (b) The rectification is produced using the circuit of Fig. 7.1.

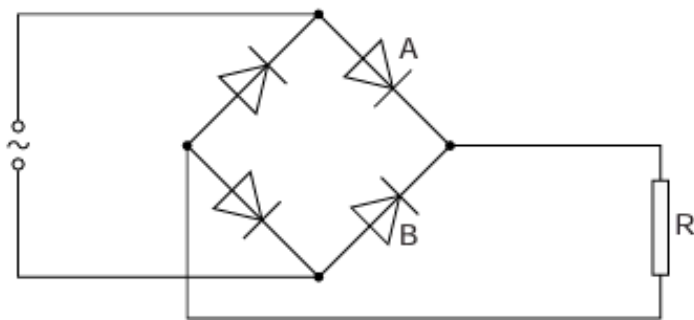


Fig. 7.1

All the diodes may be considered to be ideal.

The variation with time t of the alternating voltage applied to the circuit is shown in Fig. 7.2 and in Fig. 7.3.

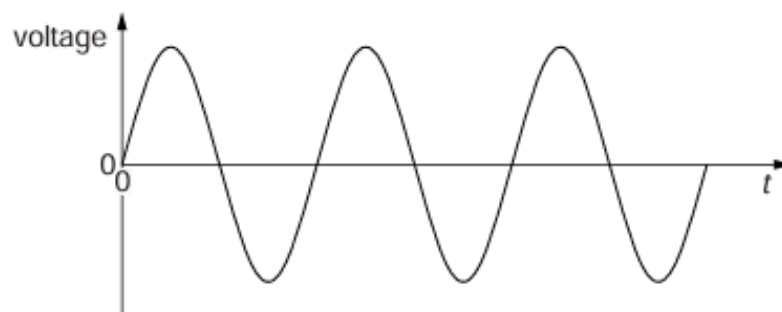
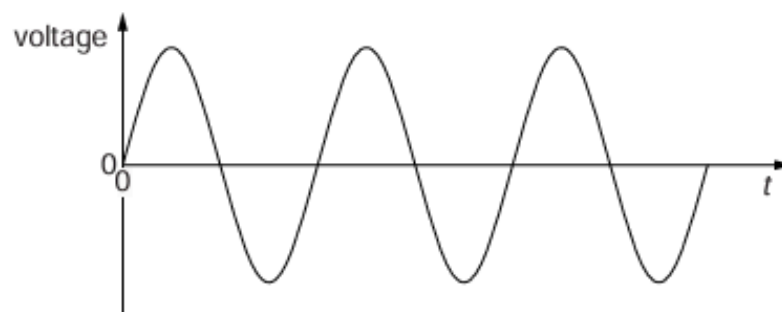


Fig. 7.2



- (i) On the axes of Fig. 7.2, draw a graph to show the variation with time t of the potential difference across diode A. [1]
- (ii) On the axes of Fig. 7.3, draw a graph to show the variation with time t of the potential difference across diode B. [1]
- (c) (i) On Fig. 7.1, draw the symbol for a capacitor, connected into the circuit so as to provide smoothing. [1]
- (ii) Fig. 7.4 shows the variation with time t of the smoothed potential difference across the resistor R in Fig. 7.1.

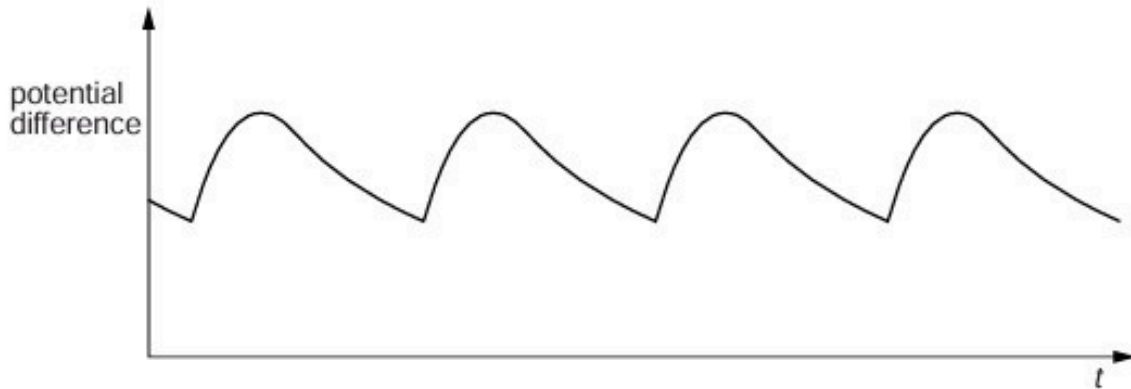


Fig. 7.4

1. State how the amount of smoothing may be increased.

.....
 [1]

2. On Fig. 7.4, draw the variation with time t of the potential difference across resistor R for increased smoothing. [2]

2. M/J/10/42/Q7

- (a) Explain what is meant by the *root-mean-square* (r.m.s.) value of an alternating voltage.

.....
.....
..... [2]

- (b) An alternating voltage V is represented by the equation

$$V = 220 \sin(120\pi t),$$

where V is measured in volts and t is in seconds.

For this alternating voltage, determine

- (i) the peak voltage,

peak voltage = V [1]

- (ii) the r.m.s. voltage,

r.m.s. voltage = V [1]

- (iii) the frequency.

frequency = Hz [1]

- (c) The alternating voltage in (b) is applied across a resistor such that the mean power output from the resistor is 1.5 kW.

Calculate the resistance of the resistor.

resistance = Ω [2]

3. O/N/10/43/Q6

The variation with time t of the current I in a resistor is shown in Fig. 6.1.

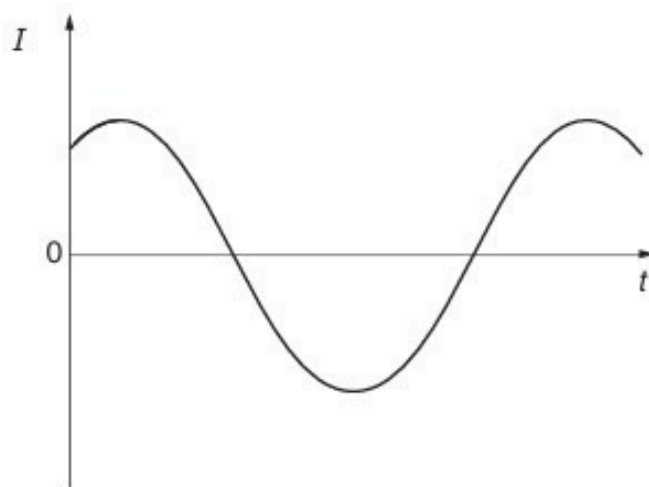


Fig. 6.1

The variation of the current with time is sinusoidal.

- (a) Explain why, although the current is not in one direction only, power is converted in the resistor.

.....
.....
..... [2]

- (b) Using the relation between root-mean-square (r.m.s.) current and peak current, deduce the value of the ratio

$$\frac{\text{average power converted in the resistor}}{\text{maximum power converted in the resistor}}.$$

ratio = [3]

4. M/J/11/42/Q6

An alternating current supply is connected in series with a resistor R , as shown in Fig. 6.1.

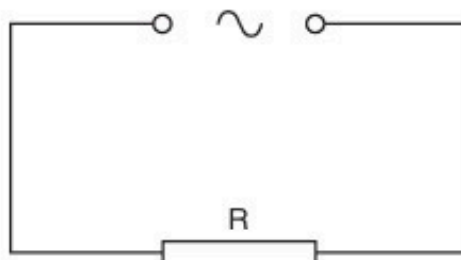


Fig. 6.1

The variation with time t (measured in seconds) of the current I (measured in amps) in the resistor is given by the expression

$$I = 9.9 \sin(380t).$$

(a) For the current in the resistor R , determine

(i) the frequency,

frequency = Hz [2]

(ii) the r.m.s. current.

r.m.s. current = A [2]

- (b) To prevent over-heating, the mean power dissipated in resistor R must not exceed 400W.
Calculate the minimum resistance of R.

resistance = Ω [2]

5. O/N/11/41/Q6

The variation with time t of the output V of an alternating voltage supply of frequency 50 Hz is shown in Fig. 6.1.

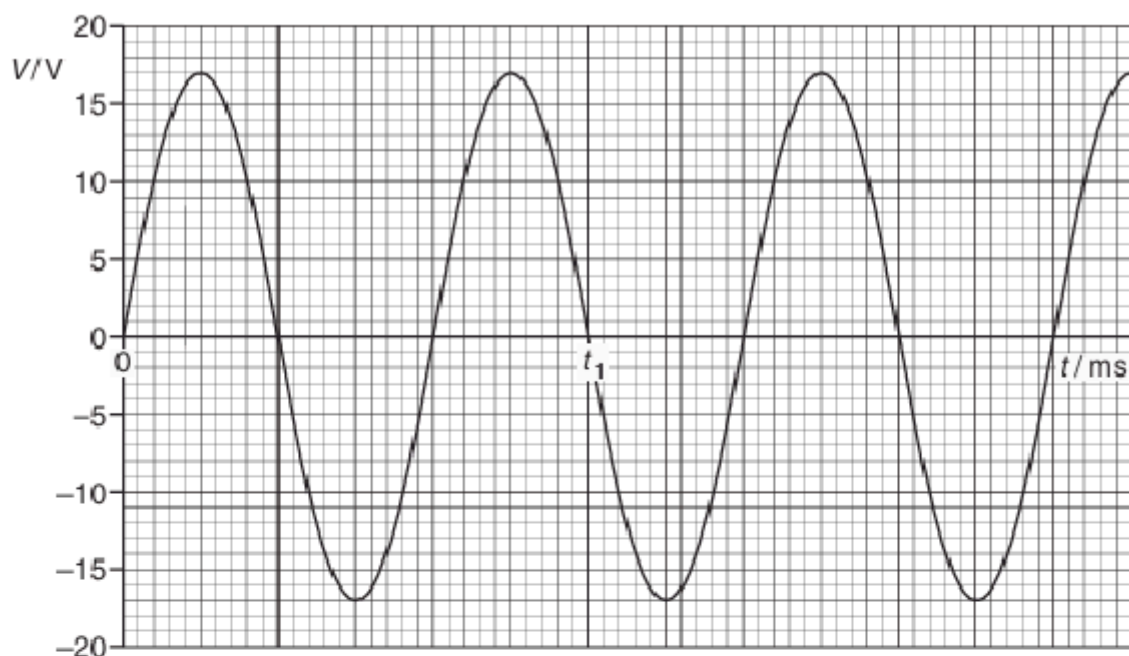


Fig. 6.1

(a) Use Fig. 6.1 to state

(i) the time t_1 ,

$$t_1 = \dots\dots\dots \text{ s [2]}$$

(ii) the peak value V_0 of the voltage,

$$V_0 = \dots\dots\dots \text{ V [1]}$$

(iii) the root-mean-square voltage V_{rms} ,

$$V_{\text{rms}} = \dots\dots\dots \text{ V [1]}$$

(iv) the mean voltage $\langle V \rangle$.

$$\langle V \rangle = \dots\dots\dots \text{ V [1]}$$

- (b) The alternating supply is connected in series with a resistor of resistance $2.4\ \Omega$. Calculate the mean power dissipated in the resistor.

power = W [2]

6. O/N/11/43/Q5

The components for a bridge rectifier are shown in Fig. 5.1.

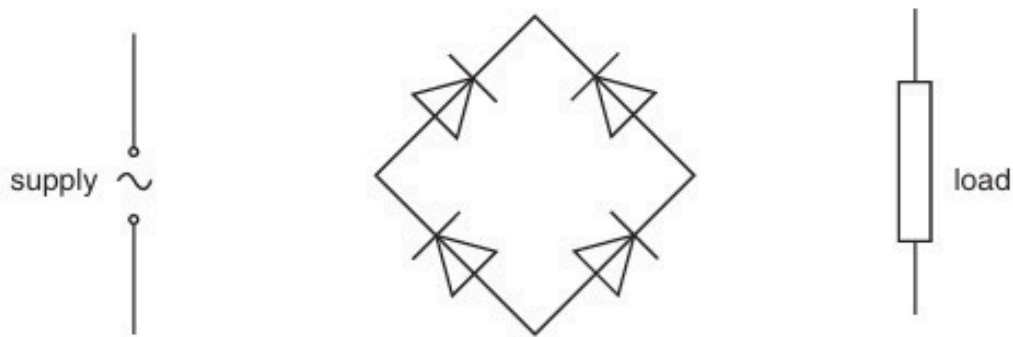


Fig. 5.1

- (a) Complete the circuit of Fig. 5.1 by showing the connections of the supply and of the load to the diodes. [2]

- (b) Suggest one advantage of the use of a bridge rectifier, rather than a single diode, for the rectification of alternating current.

.....
..... [1]

- (c) State

- (i) what is meant by *smoothing*,

.....
..... [1]

- (ii) the effect of the value of the capacitance of the smoothing capacitor in relation to smoothing.

.....
.....
..... [2]

7. M/J/12/41/Q6

A sinusoidal alternating voltage supply is connected to a bridge rectifier consisting of four ideal diodes. The output of the rectifier is connected to a resistor R and a capacitor C as shown in Fig. 6.1.

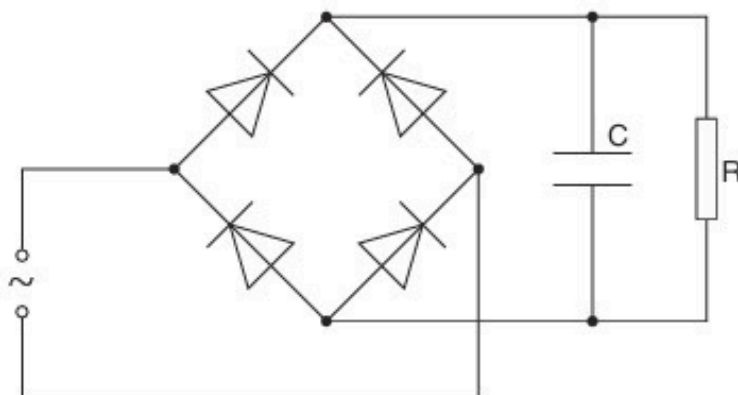


Fig. 6.1

The function of C is to provide some smoothing to the potential difference across R . The variation with time t of the potential difference V across the resistor R is shown in Fig. 6.2.

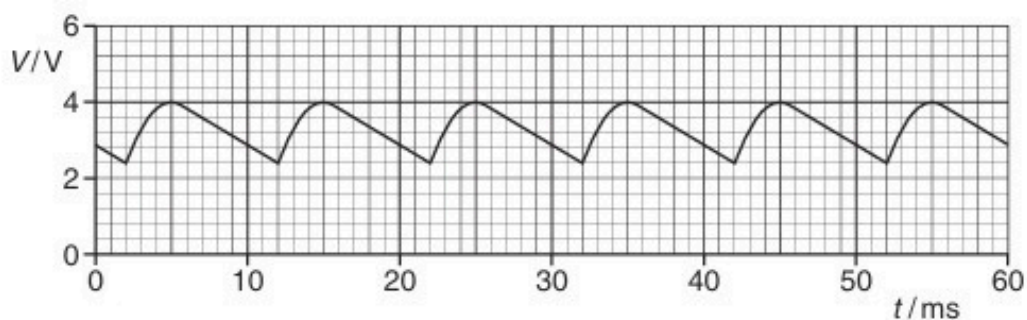


Fig. 6.2

(a) Use Fig. 6.2 to determine, for the alternating supply,

(i) the peak voltage,

peak voltage = V [1]

(ii) the root-mean-square (r.m.s.) voltage,

r.m.s. voltage = V [1]

(iii) the frequency. Show your working.

frequency = Hz [2]

(b) The capacitor C has capacitance $5.0\ \mu\text{F}$.

For a single discharge of the capacitor through the resistor R, use Fig. 6.2 to

(i) determine the change in potential difference,

change = V [1]

(ii) determine the change in charge on each plate of the capacitor,

change = C [2]

(iii) show that the average current in the resistor is $1.1 \times 10^{-3}\text{ A}$.

- (c) Use Fig. 6.2 and the value of the current given in (b)(iii) to estimate the resistance of resistor R.

resistance = Ω [2]

8. O/N/12/43/Q6

A bridge rectifier consists of four ideal diodes A, B, C and D, connected as shown in Fig. 6.1.

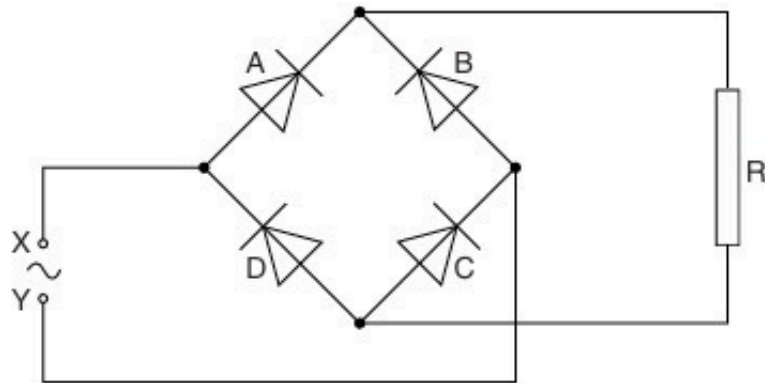


Fig. 6.1

An alternating supply is applied between the terminals X and Y.

(a) (i) On Fig. 6.1, label the positive (+) connection to the load resistor R. [1]

(ii) State which diodes are conducting when terminal Y of the supply is positive.

diode and diode [1]

(b) The variation with time t of the potential difference V across the load resistor R is shown in Fig. 6.2.

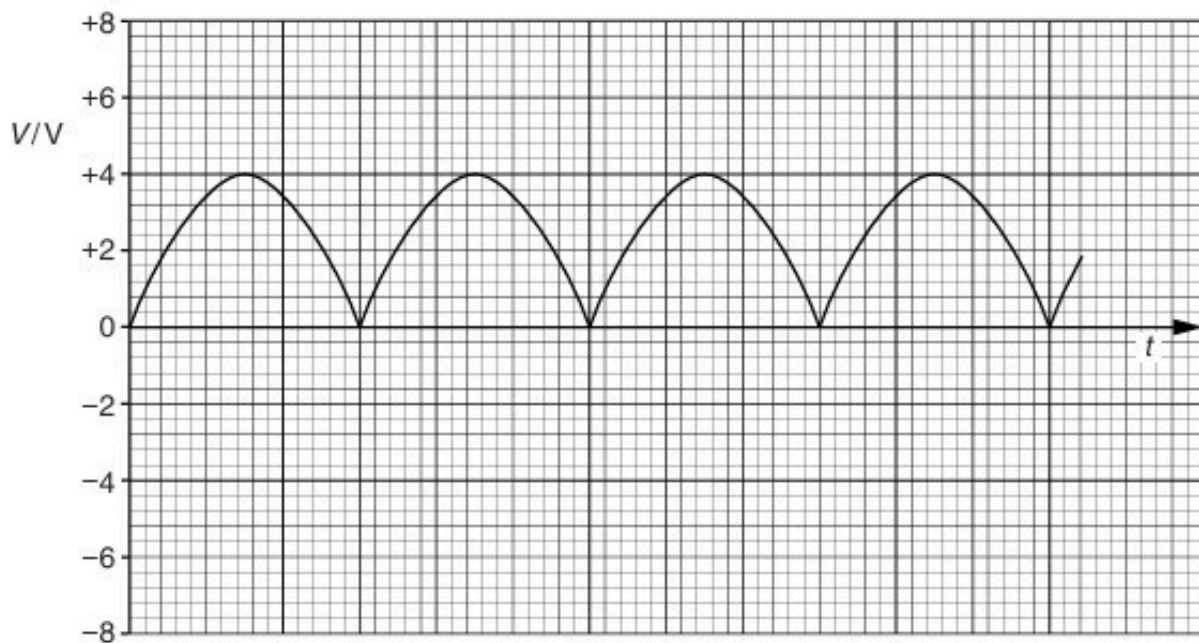


Fig. 6.2

The load resistor R has resistance $2700\ \Omega$.

- (i) Use Fig. 6.2 to determine the mean power dissipated in the resistor R.

power = W [3]

- (ii) On Fig. 6.1, draw the symbol for a capacitor, connected so as to increase the mean power dissipated in the resistor R. [1]

- (c) The capacitor in (b)(ii) is now removed from the circuit.
The diode A in Fig. 6.1 stops functioning, so that it now has infinite resistance.

On Fig. 6.2, draw the variation with time t of the new potential difference across the resistor R. [2]

9. O/N/15/43/Q7

A student is using a power supply that produces a sinusoidal output. The meters on the supply show that the output voltage V has a root-mean-square (r.m.s.) value of 14V with a frequency of 750 Hz.

The variation with time t of the output voltage V may be represented by the expression

$$V = V_0 \sin \omega t.$$

(a) Determine the value of

(i) V_0 ,

$$V_0 = \dots\dots\dots \text{V} \quad [1]$$

(ii) ω .

$$\omega = \dots\dots\dots \text{rad s}^{-1} \quad [1]$$

(b) A capacitor with a large capacitance is connected across the terminals of the supply.

Suggest and explain why this may lead to a large current from the supply.

.....
.....
..... [3]

10. M/J/16/41/Q11

The variation with time t of the sinusoidal current I in a resistor of resistance $450\,\Omega$ is shown in Fig. 11.1.

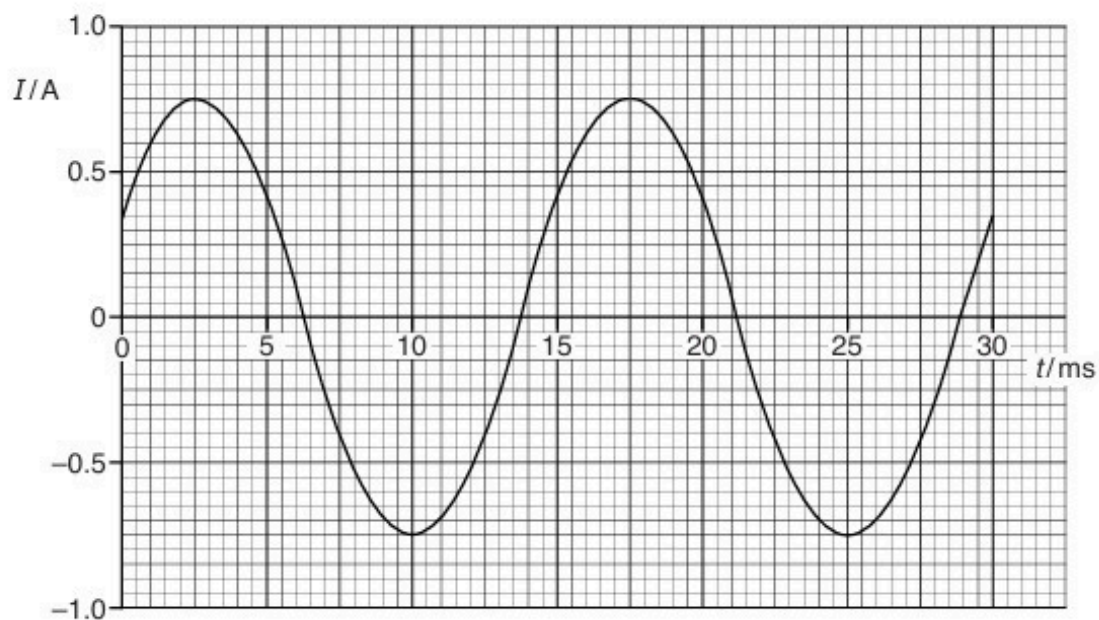


Fig. 11.1

Use data from Fig. 11.1 to determine, for the time $t = 0$ to $t = 30\text{ ms}$,

(a) the frequency of the current,

frequency = Hz [2]

(b) the mean current,

mean current = A [1]

(c) the root-mean-square (r.m.s.) current,

r.m.s. current = A [2]

(d) the energy dissipated by the resistor.

energy =J [2]

[Total: 7]

11. M/J/16/42/Q11

A bridge rectifier contains four ideal diodes A, B, C and D, as shown in Fig. 11.1.

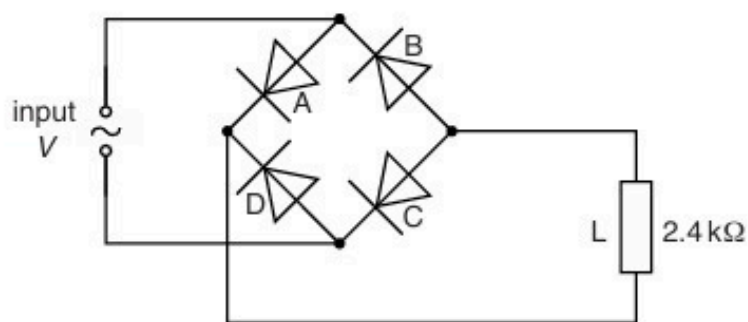


Fig. 11.1

The output of the rectifier is connected to a load L of resistance $2.4\text{ k}\Omega$.

- (a) On Fig. 11.1, mark with the letter P the positive terminal of the load. [1]
- (b) The variation with time t of the potential difference V across the input to the rectifier is shown in Fig. 11.2.

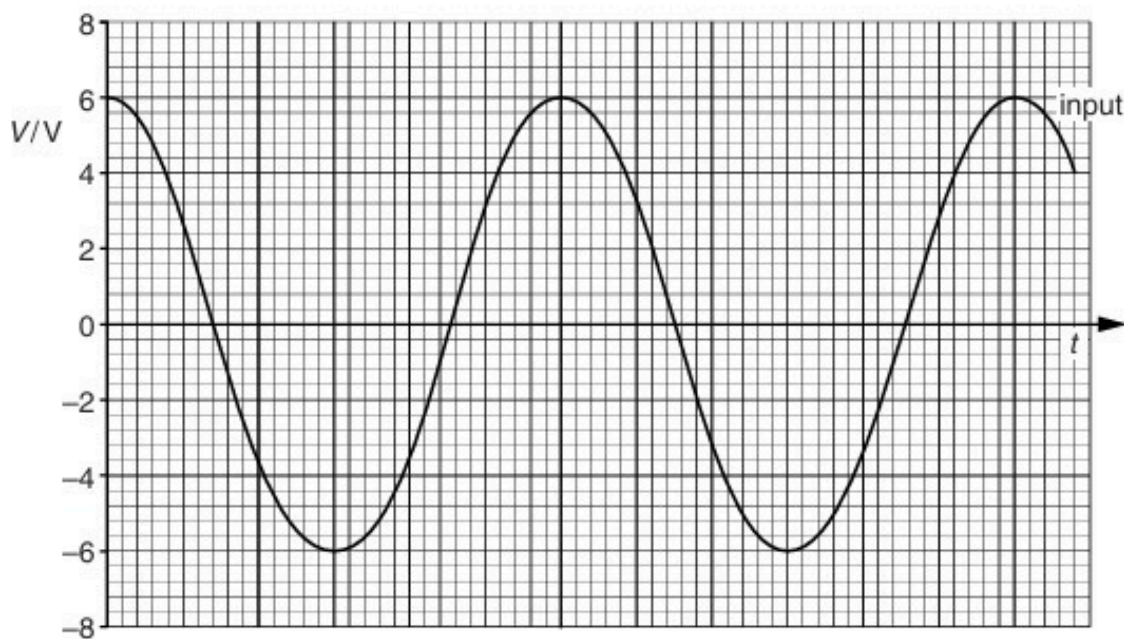


Fig. 11.2

Calculate the root-mean-square (r.m.s.) current in the load L.

(c) The potential difference across the load L is to be smoothed using a capacitor.

(i) On Fig. 11.1, draw the symbol for a capacitor, connected to produce smoothing. [1]

(ii) The minimum potential difference across the load L with the smoothing capacitor connected is 3.0 V.

On Fig. 11.2, sketch the variation with time t of the potential difference across the load L . [3]

[Total: 7]

12. O/N/17/42/Q11

The circuit for a full-wave rectifier using four ideal diodes is shown in Fig. 11.1.

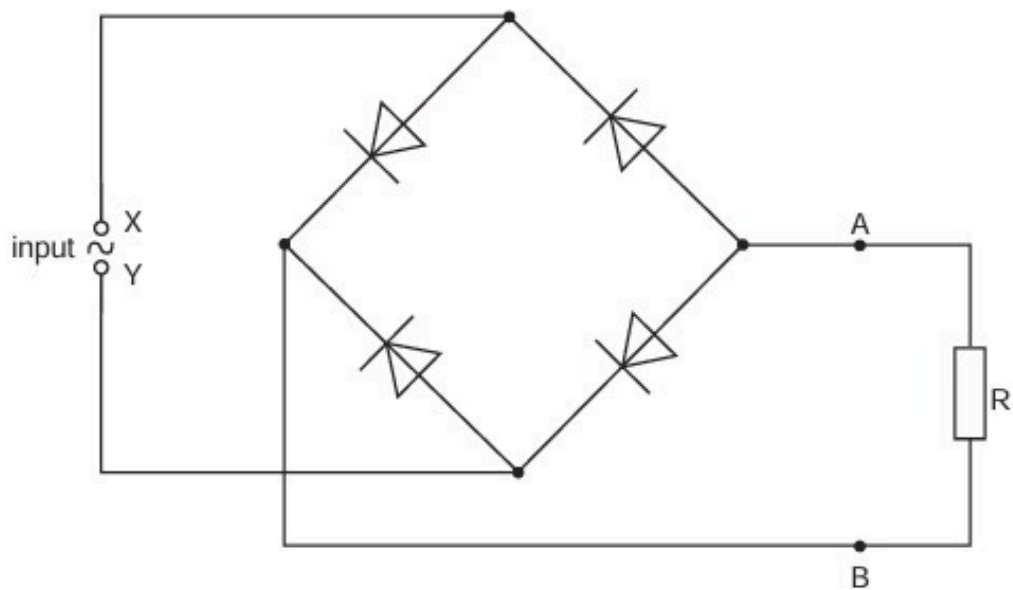


Fig. 11.1

A resistor R is connected across the output AB of the rectifier.

(a) On Fig. 11.1,

- (i)** draw a circle around any diodes that conduct when the terminal X of the input is positive with respect to terminal Y, [1]
- (ii)** label the positive (+) and the negative (–) terminals of the output AB. [1]

- (b) The variation with time t of the potential difference V across the input XY is given by the expression

$$V = 5.6 \sin 380t$$

where V is measured in volts and t is measured in seconds.

The variation with time t of the rectified potential difference across the resistor R is shown in Fig. 11.2.

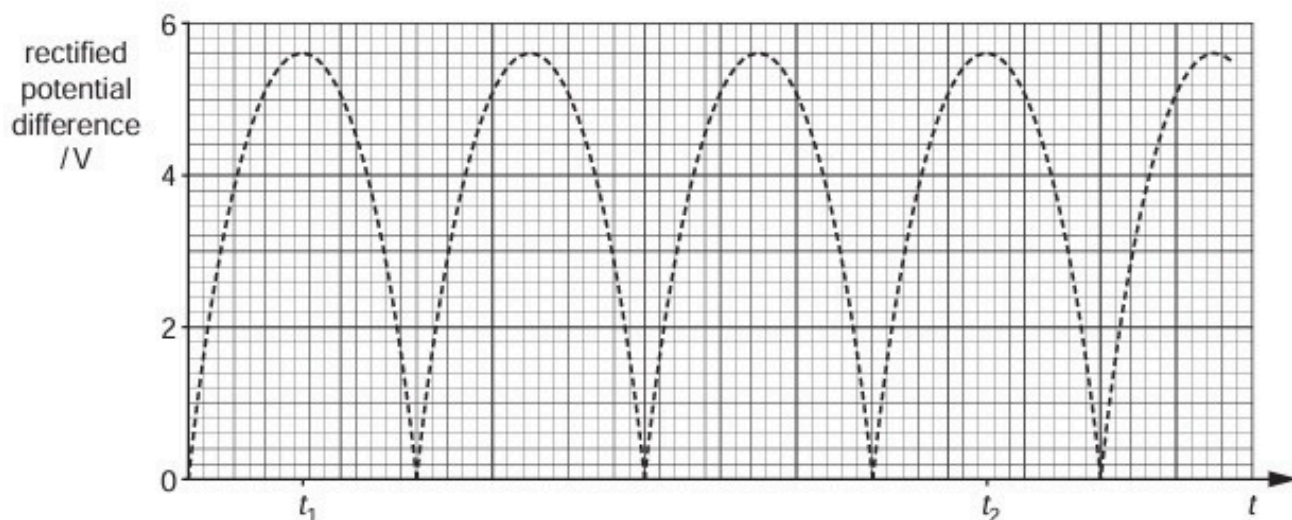


Fig. 11.2

Use the expression for the input potential difference V , or otherwise, to determine

- (i) the root-mean-square (r.m.s.) potential difference $V_{\text{r.m.s.}}$ of the input,

$$V_{\text{r.m.s.}} = \dots\dots\dots \text{ V [1]}$$

- (ii) the number of times per second that the rectified potential difference at the output reaches a peak value.

number = [2]

- (c) A capacitor is now connected between the terminals AB of the output.
The capacitor reduces the variation (the ripple) in the output to 1.6 V.

- (i) On Fig. 11.2, sketch the variation with time t of the smoothed output voltage for time $t = t_1$ to time $t = t_2$. [4]

- (ii) Suggest and explain the effect, if any, on the mean power dissipation in resistor R when the capacitor is connected between terminals AB.

.....
.....
.....[2]

[Total: 11]

13. M/J/19/41/Q10

A bridge rectifier contains four diodes. The output of the rectifier is connected to a resistor R , as shown in Fig. 10.1.

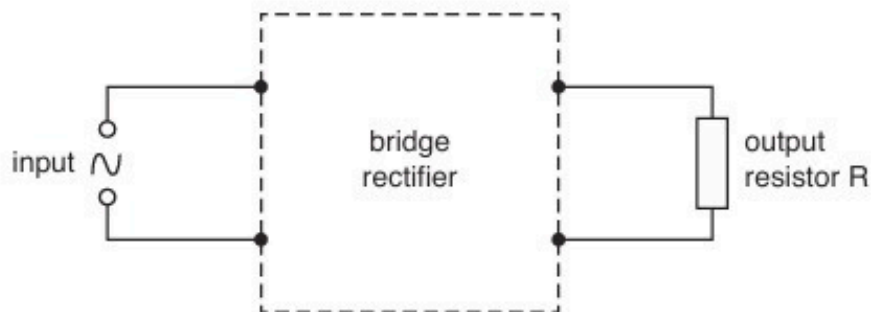


Fig. 10.1

The variation with time t of the input e.m.f. E to the rectifier is given by the expression

$$E = 15 \cos(210t)$$

where t is measured in seconds and E in volts.

The variation with time t of the potential difference V across resistor R is shown in Fig. 10.2.

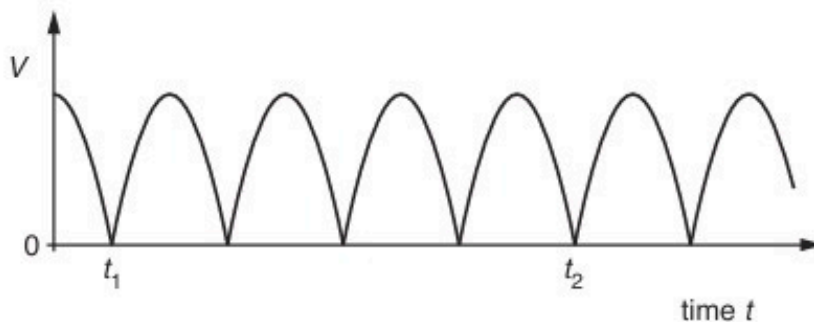


Fig. 10.2

Determine:

- (a) the maximum potential difference V_{MAX} across resistor R

$$V_{\text{MAX}} = \dots\dots\dots \text{V [1]}$$

(b) the time interval, to two significant figures, between time t_1 and time t_2 .

time = s [3]

[Total: 4]

14. O/N/19/41/Q10

A bridge rectifier using four ideal diodes is shown in Fig. 10.1.

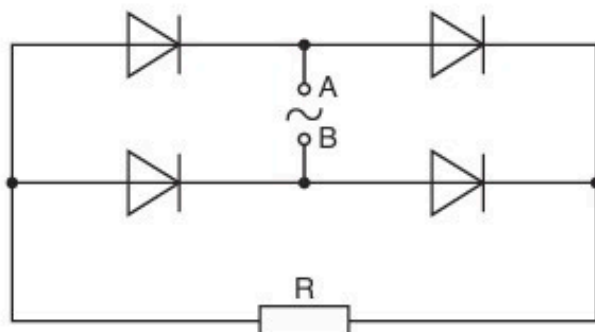


Fig. 10.1

The sinusoidal alternating electromotive force (e.m.f.) applied between points A and B has a root-mean-square (r.m.s.) value of 7.0 V.

- (a) (i) On Fig. 10.1, circle the diodes that conduct when point B is positive with respect to point A. [1]
- (ii) Calculate the maximum potential difference V_{MAX} across resistor R.

$V_{\text{MAX}} = \dots\dots\dots \text{ V [1]}$

- (b) A capacitor is connected into the circuit to produce smoothing of the potential difference across resistor R.

The variation with time t of the potential difference V across resistor R is shown in Fig. 10.2.

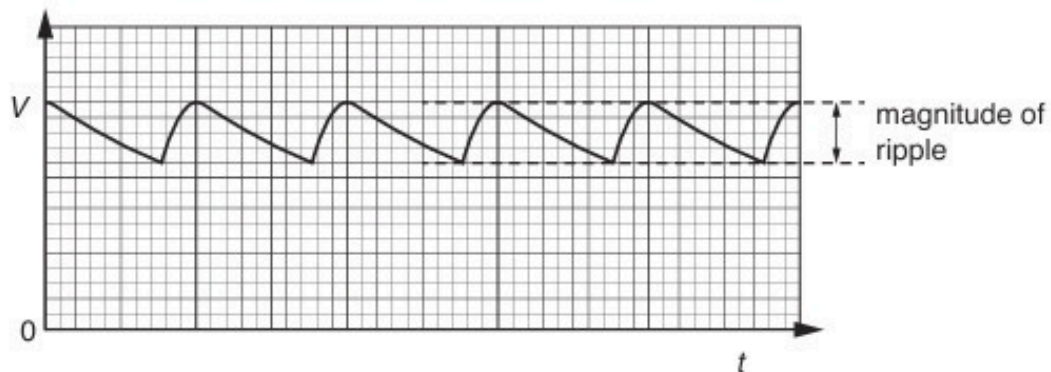


Fig. 10.2

- (i) On Fig. 10.1, draw the symbol for a capacitor, connected so as to produce smoothing. [1]

- (ii) State the effect, if any, on the magnitude of the ripple on V when, separately:

1. a capacitor of larger capacitance is used

.....

2. the resistor R has a smaller resistance.

.....

[2]

[Total: 5]

15. F/M/21/42/Q10

The output potential difference (p.d.) of an alternating power supply is represented by

$$V = 320 \sin(100\pi t)$$

where V is the p.d. in volts and t is the time in seconds.

- (a) Determine the root-mean-square (r.m.s.) p.d. of the power supply.

r.m.s. p.d. = V [1]

- (b) Determine the period T of the output.

$T =$ s [2]

- (c) The power supply is connected to resistor R and a diode in the circuit shown in Fig. 10.1.

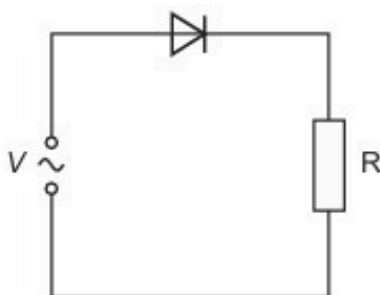


Fig. 10.1

- (i) State the name of the type of rectification produced by the diode in Fig. 10.1.

..... [1]

- (ii) On Fig. 10.2 sketch the variation with time t of the p.d. V_R across R from time $t = 0$ to time $t = 40$ ms.

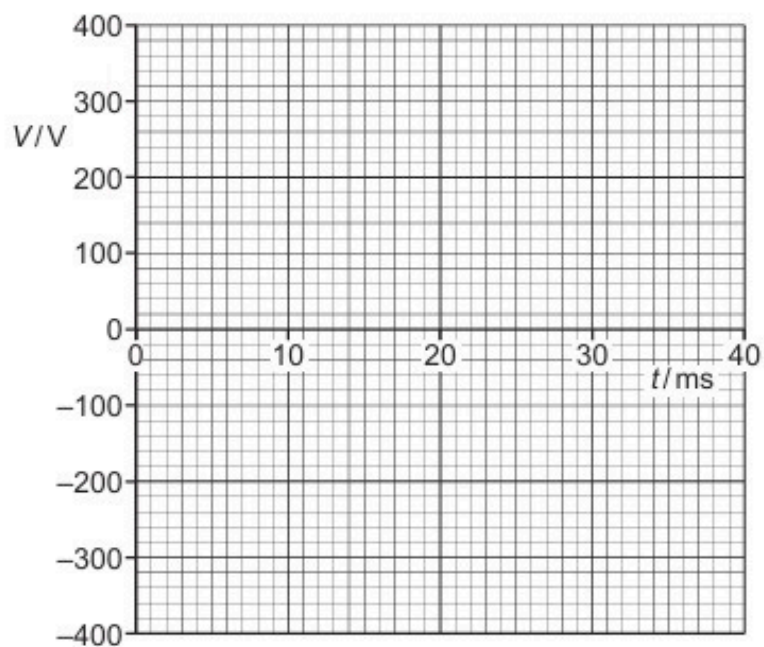


Fig. 10.2

[3]

- (iii) On Fig. 10.1, draw the symbol for a component that may be connected to produce smoothing of V_R .

[1]

[Total: 8]

16. M/J/21/42/Q10

- (a) By reference to heating effect, explain what is meant by the *root-mean-square (r.m.s.)* value of an alternating current.

.....
.....
..... [2]

- (b) The variations with time t of two currents I_1 and I_2 are shown in Fig. 10.1 and Fig. 10.2.

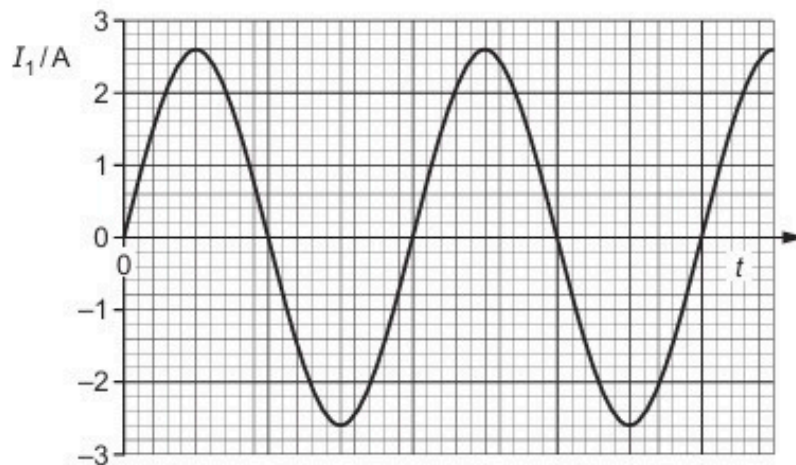


Fig. 10.1

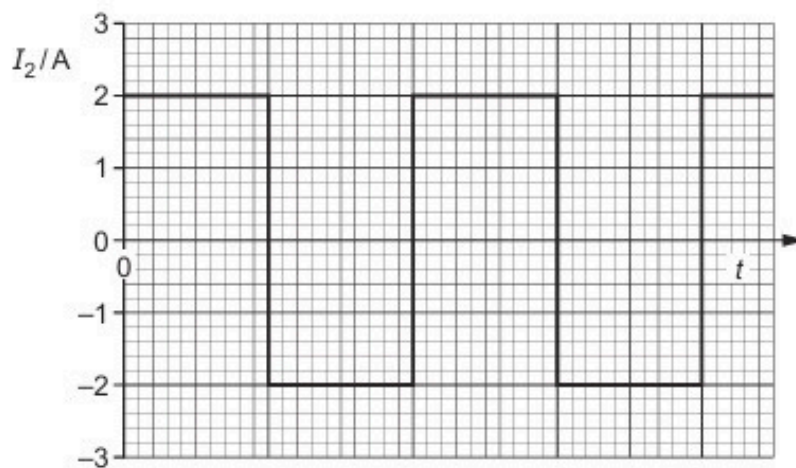


Fig. 10.2

- (i) Use Fig. 10.1 to determine the peak value and the r.m.s. value of the current I_1 .

peak value = A

r.m.s. value = A
[1]

- (ii) Use Fig. 10.2 to determine the peak value and the r.m.s. value of the current I_2 .

peak value = A

r.m.s. value = A
[1]

- (c) The variation with time t of the supply voltage V to a house is given by the expression

$$V = 240 \sin kt$$

where V is in volts, t is in seconds and k is a constant with unit rad s^{-1} .

- (i) The frequency of the supply voltage is 50 Hz.

Determine k to two significant figures.

$k = \dots\dots\dots \text{rad s}^{-1}$ [2]

- (ii) The supply voltage is applied to a heater. The mean power of the heater is 3.2 kW.

Calculate the resistance of the heater.

resistance = Ω [2]

[Total: 8]

17. O/N/21/41/Q9

- (a) State, by reference to the power dissipated in a resistor, what is meant by the *root-mean-square (r.m.s.)* value of an alternating voltage.

.....

 [2]

- (b) A coil is rotating freely, on frictionless bearings, at constant speed in a uniform magnetic field. This rotation causes an induced alternating electromotive force (e.m.f.) across the open terminals of the coil. The induced e.m.f. has r.m.s. value 12 V and frequency 50 Hz.

The speed of rotation of the coil is now doubled.

- (i) State and explain, with reference to the principles of electromagnetic induction, the effect of the increased speed of rotation on the r.m.s. value of the induced e.m.f.

.....

 [2]

- (ii) On Fig. 9.1, sketch the variation with time t of the induced e.m.f. E across the terminals of the coil at the **increased** speed of rotation. Your line should extend from time $t = 0$ to time $t = 20$ ms. Assume that $E = 0$ when $t = 0$.

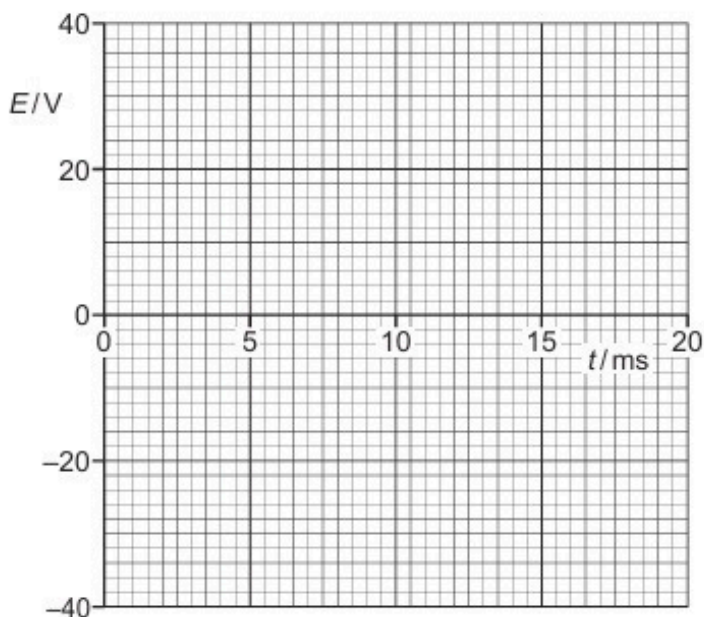


Fig. 9.1

- (c) State and explain the effect on the motion of the coil in (b) of connecting a load resistor across its terminals.

.....

.....

.....

..... [2]

[Total: 9]

18. F/M/22/42/Q7

- (a) Alternating current (a.c.) is converted into direct current (d.c.) using a full-wave rectification circuit. Part of the diagram of this circuit is shown in Fig. 7.1.

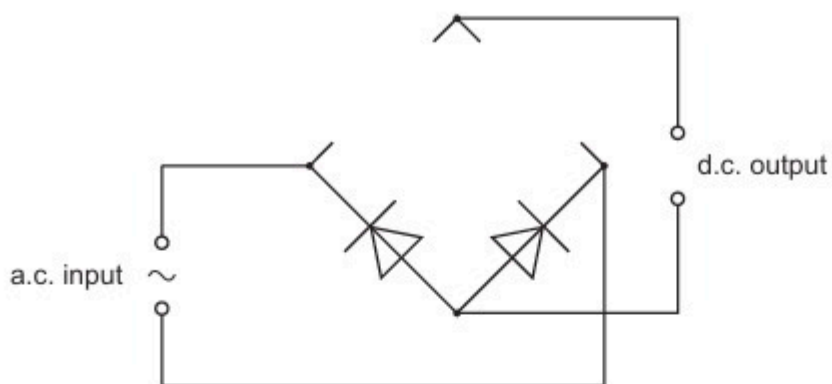


Fig. 7.1

- (i) Complete the circuit in Fig. 7.1 by adding the necessary components in the gaps. [1]
- (ii) On Fig. 7.1 mark with a + the positive output terminal of the rectifier. [1]
- (b) The output voltage V of an a.c. power supply varies sinusoidally with time t as shown in Fig. 7.2.

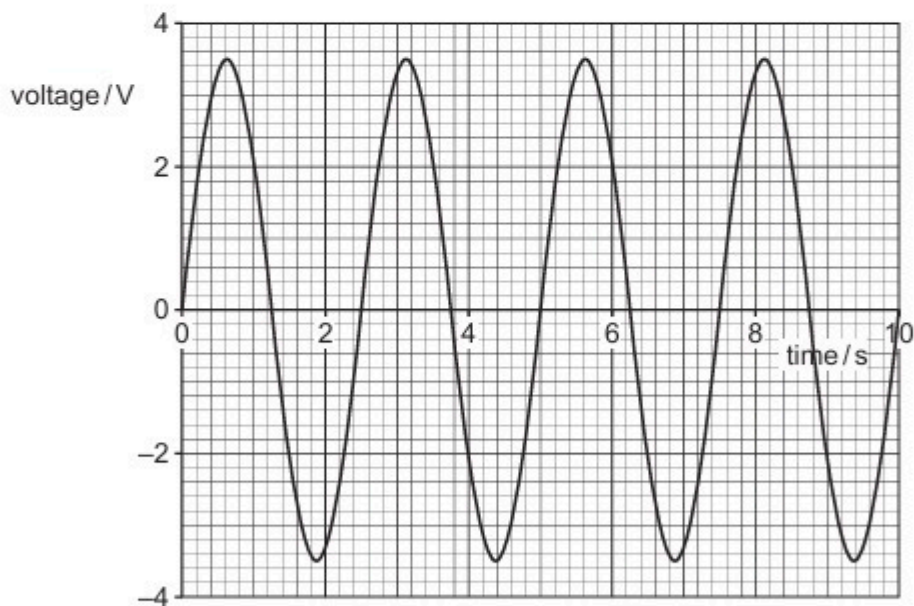


Fig. 7.2

- (i) Determine the equation for V in terms of t , where V is in volts and t is in seconds.

$V = \dots\dots\dots$ [2]

- (ii) The supply is connected to a $12\ \Omega$ resistor. Calculate the mean power dissipated in the resistor.

mean power = W [2]

[Total: 6]

19. M/J/22/41/Q5

Fig. 5.1 shows four diodes and a load resistor of resistance $1.2\text{ k}\Omega$, connected in a circuit that is used to produce rectification of an alternating voltage.

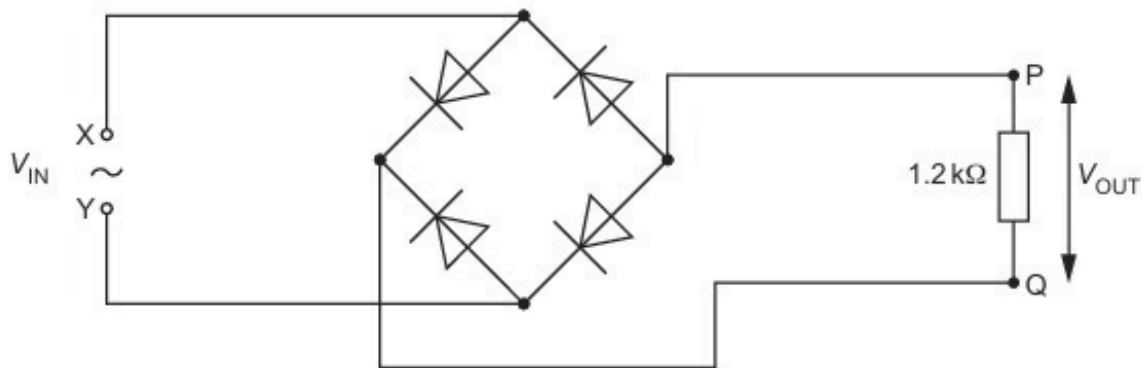


Fig. 5.1

- (a) (i)** State what is meant by rectification.

.....
..... [1]

- (ii)** State the type of rectification produced by the circuit in Fig. 5.1.

..... [1]

- (b)** A sinusoidal alternating voltage V_{IN} is applied across the input terminals X and Y. The variation with time t of V_{IN} is given by the equation

$$V_{IN} = 6.0 \sin 25\pi t$$

where V_{IN} is in volts and t is in seconds.

- (i)** On Fig. 5.1, label the output terminals P and Q with the appropriate symbols to indicate the polarity of the output voltage V_{OUT} . [1]

- (ii) The magnitude of the output voltage V_{OUT} varies with t as shown in Fig. 5.2.

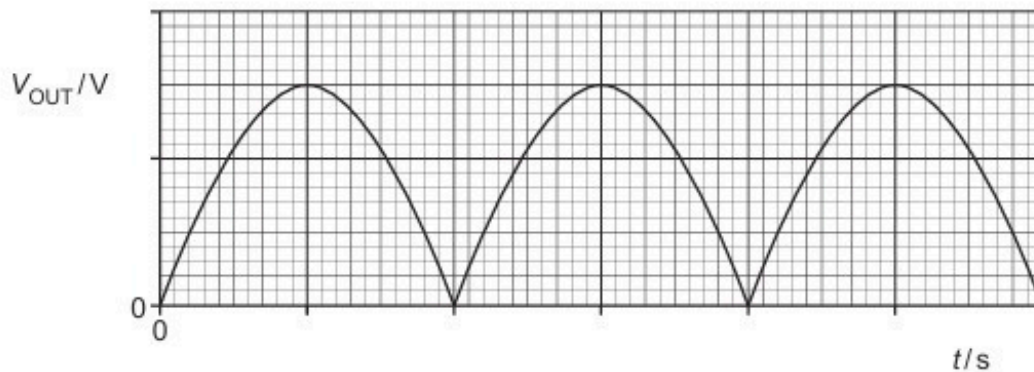


Fig. 5.2

On Fig. 5.2, label both of the axes with the correct scales. Use the space below for any working that you need.

[3]

- (c) The output voltage in (b) is smoothed by adding a capacitor to the circuit in Fig. 5.1.
The difference between the maximum and minimum values of the smoothed output voltage is 10% of the peak voltage.

- (i) On Fig. 5.1, draw the circuit symbol for a capacitor showing the capacitor correctly connected into the circuit. [1]

- (ii) On Fig. 5.2, sketch the variation with t of the smoothed output voltage. [2]

- (iii) Calculate the capacitance C of the capacitor.

$C = \dots\dots\dots \text{ F [3]}$

[Total: 12]

20. O/N/22/41/Q7

- (a) A sinusoidal alternating voltage has a root-mean-square (r.m.s.) potential difference (p.d.) of 4.2 V and a frequency of 50 kHz.

- (i) The alternating voltage is applied across a resistor of resistance 760 Ω .

By considering the peak voltage, show that the maximum power dissipated by the resistor is 46 mW.

[2]

- (ii) On Fig. 7.1, draw a smooth curve to show how the power P dissipated in the resistor varies with time t between $t = 0$ and $t = 40 \mu\text{s}$. Assume that $P = 0$ when $t = 0$.

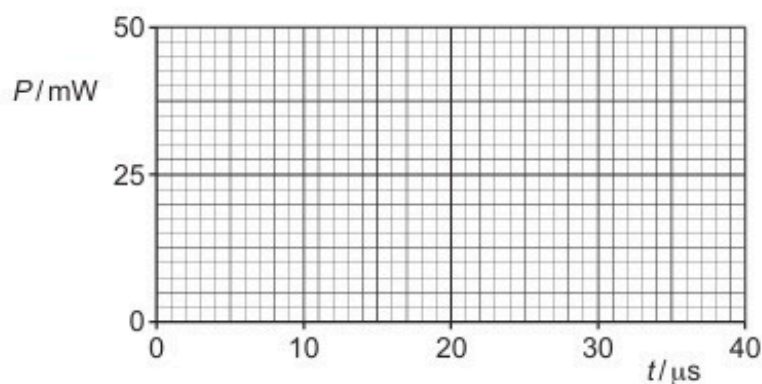


Fig. 7.1

[3]

- (iii) Use your line in (a)(ii) to explain why the mean power dissipated in the resistor is 23 mW.

.....

 [1]

(b) The alternating voltage in **(a)** is now applied to a piezoelectric crystal in air.

(i) Explain what happens to the air surrounding the crystal.

.....

.....

.....

..... [3]

(ii) A second piezoelectric crystal is placed in the air near to the first crystal.

Explain the effect of the surrounding air in **(b)(i)** on the second crystal.

..... [1]

[Total: 10]

21. M/J/23/41/Q5

Part of an electric circuit is shown in Fig. 5.1.

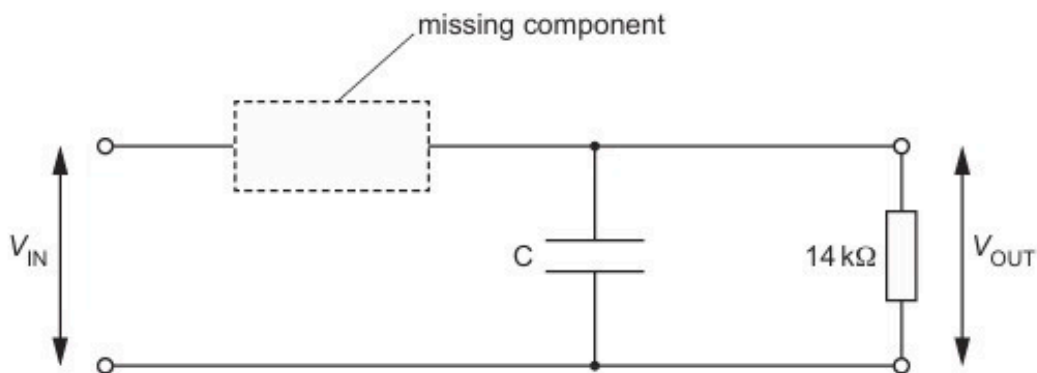


Fig. 5.1

The circuit is used to produce half-wave rectification of an alternating voltage of potential difference (p.d.) V_{IN} .

The output p.d. across the $14\text{ k}\Omega$ resistor is V_{OUT} .

- (a) (i)** A component is missing from the circuit of Fig. 5.1.

Complete the circuit diagram in Fig. 5.1 by adding the circuit symbol for the missing component, correctly connected. [1]

- (ii)** A capacitor C is shown in the circuit of Fig. 5.1.

State the effect on V_{OUT} of including the capacitor in the circuit.

..... [1]

- (b)** Fig. 5.2 shows the variation with time t of V_{IN} .

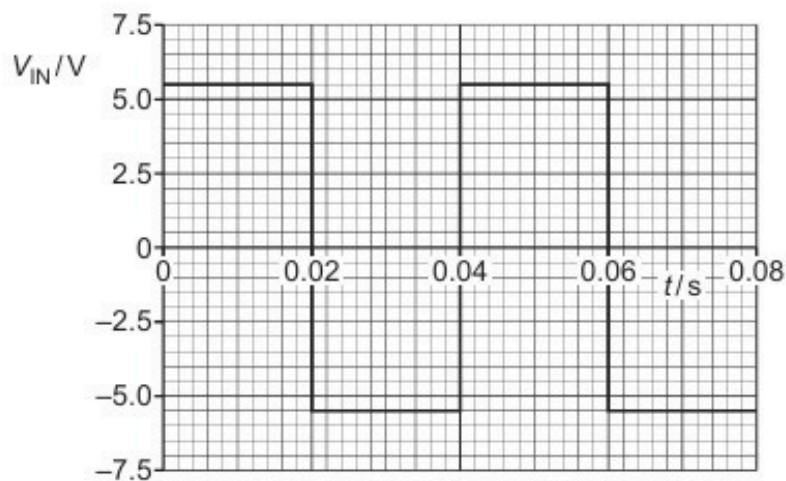


Fig. 5.2

Fig. 5.3 shows the variation with t of V_{OUT} .

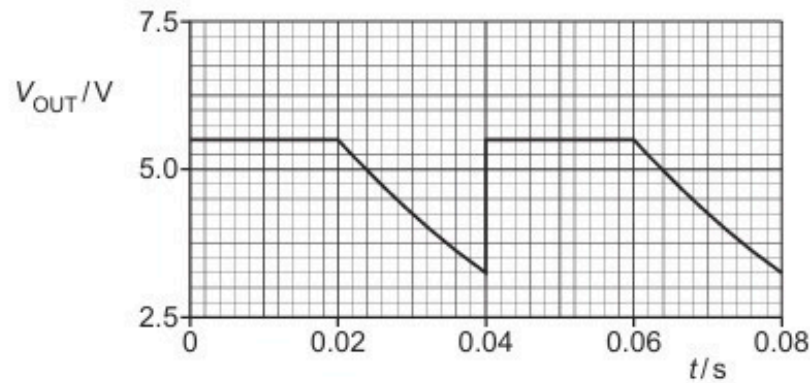


Fig. 5.3

- (i) Determine the frequency of V_{IN} .

frequency = Hz [1]

- (ii) Show that the time constant τ for the discharge of the capacitor through the resistor is 0.038 s.

[2]

- (iii) Calculate the capacitance of C. Give a unit with your answer.

capacitance = unit [2]

- (c) The circuit of Fig. 5.1 is modified so that it produces full-wave rectification of an input voltage. Suggest, with a reason, how V_{OUT} now varies with time when V_{IN} is as shown in Fig. 5.2.

.....

 [2]

22. M/J/23/42/Q7

Four diodes are used in a bridge rectifier circuit to produce rectification of a sinusoidal a.c. input voltage V_{IN} .

Fig. 7.1 shows part of the circuit, but three of the diodes are missing.

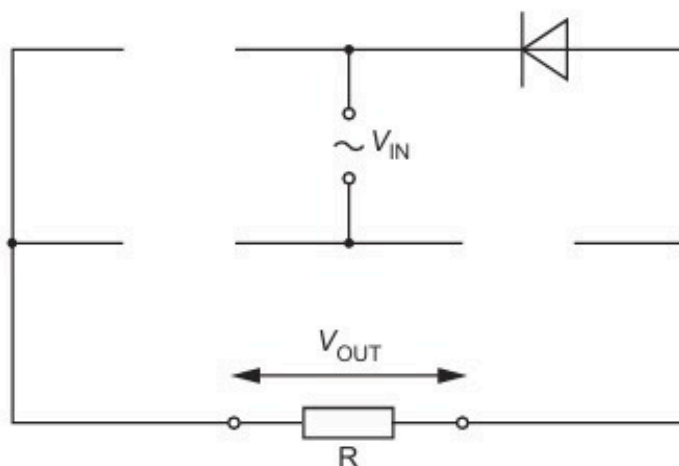


Fig. 7.1

The p.d. across the load resistor R is the output p.d. V_{OUT} of the bridge rectifier.

(a) (i) State the name of the type of rectification produced by a bridge rectifier.

..... [1]

(ii) Complete Fig. 7.1 by drawing the three missing diodes, correctly connected. [2]

(iii) On Fig. 7.1, draw an arrow to indicate the direction of the current in resistor R . [1]

(b) V_{IN} has amplitude V_0 and period T . Fig. 7.2 shows the variation with time t of V_{IN} .

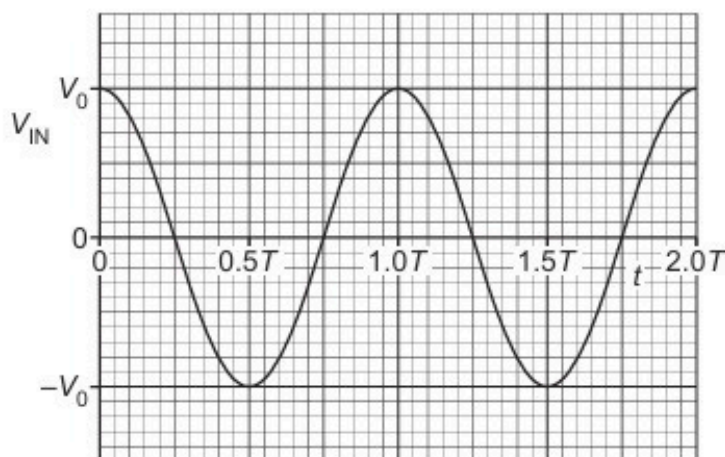


Fig. 7.2

- (i) On Fig. 7.3, sketch the variation of V_{OUT} with t between $t = 0$ and $t = 2.0T$.

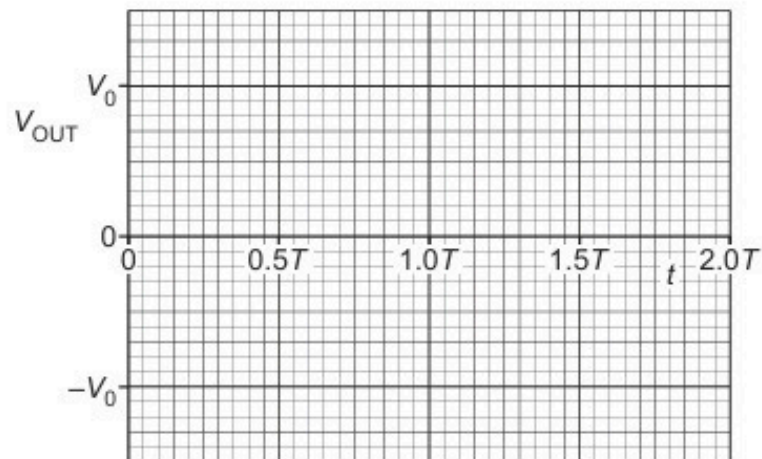


Fig. 7.3

[3]

- (ii) The power dissipated in the resistor is P .

On Fig. 7.4, sketch the variation of P with t between $t = 0$ and $t = 2.0T$.

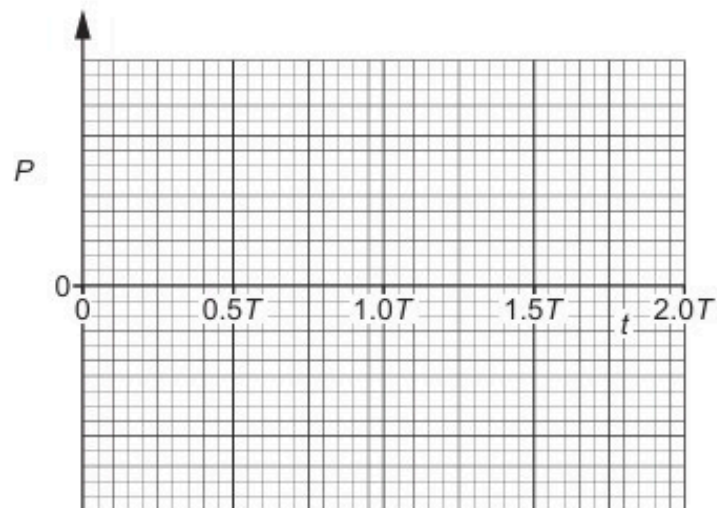


Fig. 7.4

[2]

- (iii) Suggest, with a reason, how the root-mean-square (r.m.s.) value of V_{OUT} compares with the r.m.s. value of V_{IN} .

.....

..... [1]

[Total: 10]

23. O/N/23/41/Q7

A varying current I passes through a resistor of resistance R in the circuit shown in Fig. 7.1.

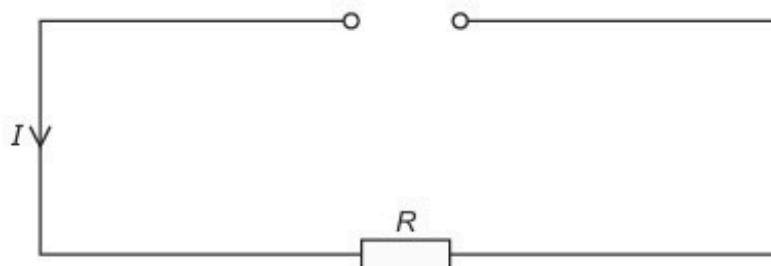


Fig. 7.1

Fig. 7.2 shows the variation with time t of I .

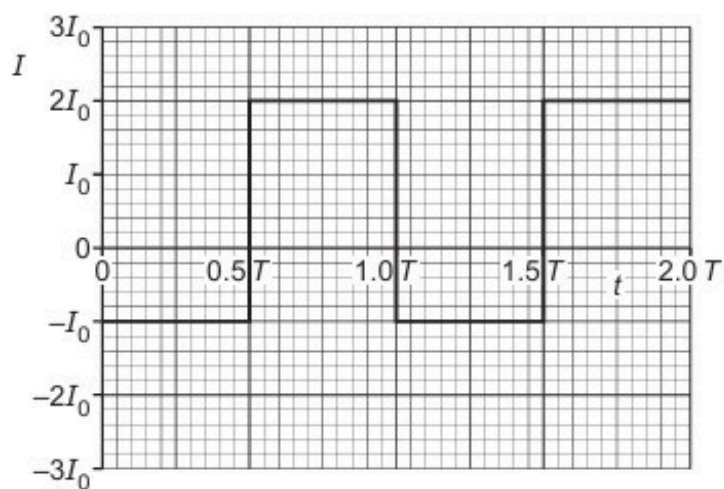


Fig. 7.2

The current has magnitude $2I_0$ when it is in the positive direction and I_0 when it is in the negative direction. The period of the variation of the current is T .

- (a) Determine expressions, in terms of I_0 and R , for the power P dissipated in the resistor for the times when:

- (i) the current is in the negative direction

$P = \dots\dots\dots [1]$

- (ii) the current is in the positive direction.

$P = \dots\dots\dots [1]$

- (b) On Fig. 7.3, sketch the variation of P with t between $t = 0$ and $t = 2.0T$. Label the power axis with an appropriate scale.

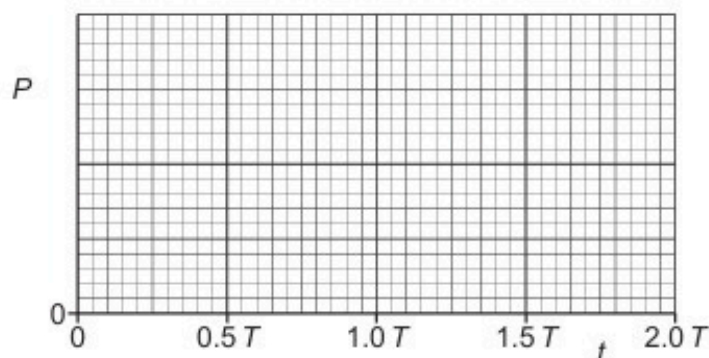


Fig. 7.3

[3]

- (c) Use your answer in (b) to determine an expression, in terms of I_0 and R , for:

- (i) the mean power $\langle P \rangle$ in the resistor

$$\langle P \rangle = \dots\dots\dots [1]$$

- (ii) the root-mean-square (r.m.s.) current $I_{\text{r.m.s.}}$ in the resistor.

$$I_{\text{r.m.s.}} = \dots\dots\dots [2]$$

[Total: 8]